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RESEARCH ARTICLE

A Two-Stage Spatiotemporal CNN-YOLOv9 Framework for High-Precision Real-Time Abandoned Object Detection in Public Surveillance Videos

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ABSTRACT Maintaining security is of prime importance in public spaces such as markets, train stations, and airports. Such situations demand reliable and advanced automated surveillance systems. This research paper presents an intelligent video surveillance system for **abandoned object detection** in public environments. The overall architecture of the proposed systems consist of two stages. In the first stage, **Convolutional Neural Networks (CNN)** is used for **real-time suspicious object recognition**, efficiently classifying detected objects into suspicious and non-suspicious categories. In the second stage, the You Only Look Once (YOLOv9) algorithm, enhanced with a **Kalman Filter for object tracking**, performs precise spatial detection, and temporal persistence analysis is employed for **abandoned object detection**. Extensive experimental results on the PETS2006 and ABODA datasets—split into 70% training, 15% validation, and 15% testing—demonstrate outstanding performance, achieving up to 99.81% accuracy, 99.67% precision, and 99.01% recall, on benchmark datasets, with potential variations in unconstrained real-world deployments. The proposed approach significantly advances **video-based threat detection** and **automated security monitoring systems**, offering high detection accuracy, robust real-time performance, and strong adaptability to complex surveillance environments.

INDEX TERMS Abandoned object detection in surveillance, automated security monitoring systems, convolutional neural networks, intelligent video surveillance, real-time suspicious object recognition, YOLOv9 object detection, Kalman filter object tracking, video-based threat detection.

I. INTRODUCTION AND BACKGROUND

A. AUTOMATED VIDEO SURVEILLANCE

Automatic Surveillance Systems are ubiquitous in modern societies in places like train stations, bus stations, airports, shopping malls, business centres and offices. Abandoned objects such as bags or packages, in such public spaces poses a serious potential threat to human lives as they might carry harmful goods like explosives and other unlawful substances. Therefore the early detection of such abandoned objects allow law enforcing agencies to respond in a timely manner to save precious lives and to avert potential destruction. Automated surveillance systems have been designed to observe human activity at in real-time and prevent acts like theft, terrorism, vandalism and accidents. Traditionally, these systems consist of a network of CCTV cameras, sensors, storage units, and control centers where human operators or automated algorithms analyze live footage in

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TABLE 1. Comparison of YOLOv9 Variants.

Model	Description	Performance	Ideal Use Case
YOLOv9-t (Tiny)	Lightest and fastest variant	High speed, lower accuracy	Real-time applications on edge devices
YOLOv9-s (Small)	Balanced speed and accuracy	Moderate accuracy, faster inference	Low-resource environments
YOLOv9-m (Medium)	Middle ground between speed and precision	Balanced speed and accuracy	General-purpose tasks
YOLOv9-c (Compact)	Enhanced accuracy with moderate resource usage	Higher accuracy, moderate speed	Precision tasks without heavy computation
YOLOv9-e (Extra Large)	Highest accuracy, highest resource demand	High precision, slower inference	Critical precision applications

real time. Traditional surveillance relies heavily on manual monitoring of live footage by human operators to detect suspicious behavior. This approach is both time-consuming and prone to human error due to physical fatigue of human operators and constant monitoring, especially when managing multiple video streams over extended periods of time. In order to alleviate these drawbacks automatic processing of surveillance videos provides a promising solution in this context. Automated detection systems reduce the need for constant human monitoring by using computer vision which employs intelligent sophisticated surveillance algorithms to identify and alert authorities about suspicious activity. This not only improves the efficiency of security operations but also enhances the overall safety of public spaces [1]. However, computer vision based automated systems are limited by their own challenges such as background subtraction, failures in dynamic environments as the lighting conditions change frequently, shadowing and movements of pedestrians. Consequently, it has been proposed to combine computer vision with Machine Learning (ML) and Deep Learning (DL), to further improve the performance of automated surveillance video analysis [2], [3]. These systems aim to identify specific behaviors, objects, or events in real time without constant human supervision with improved reliability even in challenging environments [3].

B. YOU ONLY LOOK ONCE VERSION 9 (YOLOv9)

You Only Look Once version 9 (YOLOv9) [4], [5] released on February 21, 2024, is a deep learning model architecture for object detection. Although YOLOv9 shares the same underlying architecture with its predecessors, it has several significant improvements over the previous versions. A significant upgrade is an updated Neural Network framework that uses both Feature Pyramid Network (FPN) and Path Aggregation Network (PAN), and the development of a better labeling tool that simplifies the annotation process. The FPN works uses progressively reduces spatial resolution of the input image and at the same time, increases the number of feature channels. This process results in the creation of feature maps that have the ability to detect objects with different scale and resolutions. The PAN architecture incorporates characteristics of different levels in the network through skip connections. These enhancements enables the network to capture features on different scales and resolutions. The ability is critical in the attainment of accurate object detection especially in cases involving objects that are of different sizes and shapes. The labeling tool has a variety of useful features such as automatic labeling, shortcuts to labeling, and these are also capable of customizing hot keys. A combination of these capabilities makes the annotation of images used for model training, much easier. It applies the latest deep learning methods, such as transformer-based designs, dynamic convolution, and adaptive anchor-free detection, to enhance object recognition in different settings. YOLOv9 can be used in complex scenarios with high accuracy, hence it is applicable in surveillance, autonomous driving, medical imaging and even industrial automation. The hybrid nature of YOLOv9 is one of the most important enhancements as it combines both Convolutional Neural Networks (CNNs) and vision transformers (ViTs), which enables it to better capture local and global contextual information. It also uses better feature fusion techniques which enhance better detection of small objects, objects that are occluded or that include other objects. YOLOv9 has been enhanced with new loss functions and optimization methods, which guarantee the improvement of generalization and consequently minimize false positives and high recall. When applied to real-world scenarios, the YOLOv9 has been shown to perform really well on benchmark datasets, with impressive performance in terms of mean Average Precision (mAP) and inference speed than prior versions of YOLO. The ability to process images in real time, with lower computational cost, and high accuracy in detection makes YOLOv9 a more favorable option in new generation object detection. YOLOv9 is available in a variety of versions, e.g., YOLOv9-s, YOLOv9-m, and YOLOv9-c, etc. [4]. The different variants are summarized in the Table 1.

To address the limitations of conventional automated surveillance systems, this paper presents a two-stage intelligent video surveillance framework for abandoned object detection in public environments. In the first stage, a CNN performs real-time appearance-based classification to distinguish suspicious objects from non-suspicious ones. In the second stage, YOLOv9, enhanced with Kalman filter–based object tracking, conducts precise spatial localization and temporal persistence analysis to identify abandoned objects. The proposed framework is evaluated on the PETS2006 and ABODA datasets, demonstrating effective detection performance under diverse surveillance scenarios. To the best of our knowledge, this work represents one of the first systematic integrations of CNN-based appearance filtering with YOLOv9 and Kalman-based temporal verification for real-time abandoned object detection. This paper makes the following key contributions:

1) **A Two-Stage Spatiotemporal Abandoned Object Detection Framework** We propose a novel two-stage surveillance framework that decouples appearance-based suspicious object classification from spatiotemporal abandoned object verification. A CNN is used exclusively as an early-stage appearance filter to classify candidate objects as suspicious or non-suspicious, while YOLOv9, enhanced with Kalman filtering, performs precise spatial localization and temporal tracking to determine abandonment.
2) **CNN-Guided Early Filtering for Computational Efficiency and Precision** The proposed framework introduces an appearance-based CNN pre-filter that forwards only visually suspicious objects to the YOLOv9 tracking stage. This design significantly reduces unnecessary detections, lowers computational overhead, and improves precision by eliminating visually irrelevant objects at an early stage.
3) **Temporal Persistence Modeling with Kalman-Based Tracking and Thresholding** A temporal stationarity model is incorporated using Kalman filter–based object tracking and a predefined temporal persistence threshold (τ). This mechanism ensures that only objects remaining stationary beyond τ frames are classified as abandoned, effectively suppressing false alarms caused by short-term stops, occlusions, or background motion.
4) **Comprehensive Experimental Validation on Benchmark Surveillance Datasets** The proposed method is extensively evaluated on the PETS2006 and ABODA benchmark datasets under diverse surveillance conditions. Experimental results demonstrate superior performance in terms of accuracy, precision, recall, and real-time feasibility compared to existing state-of-the-art abandoned object detection approaches.

The rest of the paper is organised as follows. A review of the related work is presented in Section II, offering context and identifying existing research gaps. The details of the dataset used in this research work, its preprocessing, and its exploratory data analysis are provided in Section III. The proposed innovative two-stage model, designed to overcome key challenges in surveillance systems, is introduced in Section IV. Detailed experimental performance evaluation of the proposed scheme is provided in Section V. Finally, Conclusions are presented in Section VI.

II. LITERATURE REVIEW

Authors in [6] presented a disruptive process for identifying abandoned objects on the roads by using Vehicular Ad-hoc Networks (VANETs) and edge AI. One of the main innovations is a coordinate based location estimation technique where a projection model is used to accurately geo-locate objects in a precise way. Claims indicate that the results are highly accurate (99.57 per cent on one group) and are indicative of how effective the system is in real-time and high-precision detection at real roads. The research work proposed in [7] integrates background matting and advanced learning algorithms in a three-stage architecture for precise object identification. The authors report that their approach outperforms conventional methods, significantly reducing false identifications while maintaining high accuracy in real-world settings. A survey study on the challenge of Unsupervised Domain Adaptation (UDA) for object detection is presented in [8]. It points out that, although deep learning detection models need extensive annotated data, they perform poorly in unlabeled target domains with different visual images. The existing techniques of domain-adaptive detection are reviewed exhaustively, and their strategies as well as limitations are discussed. Dhevanandhini et. al in [9] introduces a sophisticated video surveillance network based on a hybrid deep learning approach to detect object. The methodology includes three main steps: object segmentation with the help of a Modified Barnacles Mating Optimization (MBMO) algorithm, feature optimization with the help of a Chaotic Hummingbird Optimization (CHO) algorithm, and a Hybrid CNN-Supreme Gradient Boosting (CNN-SGboost) classifier. Measured with benchmark data, the suggested hybrid CNN-SGboost model shows a high performance ratio in comparison to the current models. A fuzzy rule-based system integrating residual dense networks to perform video enhancement, distance differencing to identify objects and persons, with a fuzzy threat evaluation module, and super-resolution to reconstruct faces, has been suggested in [10]. The research work presented in [11] used background subtraction with fuzzy integral and a hierarchical Finite State Machine to identify static objects. The objects are determined to be abandoned luggage by the YOLOv5 CNN. The authors indicate that the suggested technique outperforms conventional techniques in terms of accuracy, precision and recall.

Authors in the their work in [12] mitigate the weakness of background-subtraction based algorithms to slow absorption of foregrounds and abrupt changes in illumination, both of which favor false alarms. The authors suggest a powerful algorithm in the form of a dual background model. The main innovation is a largest-contour-based presence authentication system which allows tracking an object accurately to identify abandonment despite the absence of foreground information and change in illumination. The approach is tested on a custom dataset, PETS2006 and ABODA and it has shown to be very effective in challenging conditions such as occlusion and changing lighting. The study in [13] proposes ZERO-DCE-CB for enhanced low-light images and combines it with YOLOv7-BS augmented with BiFormer and SPD-Conv, achieving 89.9% detection accuracy and minimal missed detections on a discarded object dataset. The approach demonstrates both effective image brightening and robust object recognition, with potential for broader low-light applications. The study in [14] proposes a hybrid algorithm combining MOG2 background modeling with

YOLOv9, using YOLOv9 results to guide low-confidence targets from MOG2. The method achieves 86.7% precision and 89.6% recall, improving detection performance over single algorithms and demonstrating practical value for highway management. Authors in [15] proposes LE-DETR, a lightweight detection transformer with real-time feature extraction, Triple Fusion, and Cross-Layer Multi-Scale Interaction modules to enhance multi-scale representation. Evaluated on the Highway Abandoned Object Dataset, LE-DETR improves accuracy by 6.5%, reduces parameters by 27.1%, and lowers FLOPs by 21.1%, demonstrating efficient and accurate detection for practical highway applications. A new detection method, called JSR-Net to detect unknown objects in autonomous driving is described in [16]. A reconstruction module is trained to identify and recreate the road surface. The fundamental assumption is that improper rebuilding of the road is a pointer of abnormal regions. Semantic segmentation is used with this reconstruction error to generate anomaly scores per pixel. JSR-Net shows state-of-the-art performance in terms of four benchmarks, greatly minimizing the false positives with high precision. Authors in [17] reported new developments in automatic object detection and instance segmentation in the area of earth observations. It is concerned with the issues of object recognition under variable sensors, platforms (satellite, aerial, UAV) and geographical conditions. The survey concentrates on deep learning-based approaches, that address data and label constraints. It also examines the applications that are useful in these techniques and the future developments in the detection of earth observation objects. Ingle et. al in their work in [18] introduced a compact CNN to be used in resource-constrained hardware to tackle the problem of real-time weapon-detection in multi-camera surveillance. The objective is identification and categorization of objects likes guns and knives, differentiating them with respect to normal objects. The suggested multi class and subclass CNN detection has high precision of 97.50% in particular datasets and 90.7% in multi-view camera systems, proving its efficacy in efficient, real time detection of abnormal objects. The paper [19] introduces a CNN-based algorithm that can identify freely rotated objects of all sizes, such as objects smaller than 2×2 pixels, in remote sensing images. Its method of encoding object location and orientation in grid cells using classification, unlike axis-aligned bounding boxes and anchor-based regression, results in no anchor boxes. It proposes a representation of features of rotation that are invariant to rotation over scales to deal with complete 360 degrees rotations. At inference, bounding boxes are inferred in the least amount and by grouping similar pixel groups of the same class. Analysis of the xView and the DOTA datasets indicate consistent enhancement relative to the state-of-the-art procedures.

Automated Surveillance has been proposed to track violence using computer vision [20] in real time. Authors in [21] investigate the role of object detection as a critical component in autonomous driving systems, where it is used to perceive and identify key entities such as pedestrians, other vehicles, and traffic signs. It discusses state-of-the-art deep learning-based object detectors that enable real-time performance and examines open challenges related to their integration into autonomous vehicles. A survey study presented in [22] reviews the progress of infrastructure-based object detection and tracking systems for cooperative driving automation. It examines how roadside sensors complement on-board vehicle systems and can overcome limitations like occlusion and range limitation. It concludes by highlighting current opportunities, open challenges, and future trends in infrastructure-enhanced perception for connected vehicles. A low-cost UAV-based system for autonomous trash detection and localization has been introduced in [23]. Litter is identified in images taken during autonomous patrols by a deep CNN and the objects are geo-located on a global map with the standard UAV sensors. One imortant contribution is the development and release of annotated dataset. The system was tested on embedded hardware to ensure a trade-off between detection accuracy and processing speed with concrete guidance given on the model and hardware choice. A number of reviews of object detection models offer insights into the performance of two stage models, like R-CNN, Fast R-CNN and Faster R-CNN, and single stage models such as YOLO and SSD [24]. The paper in [25] provides a comprehensive survey of CNNs, focusing on their fundamentals, architectures, and applications in image recognition. It covers core CNN components such as layers, convolution operations, and activation functions and also reviews popular models like LeNet, AlexNet, VGG, ResNet, and InceptionNet. The paper also offers practical guidance for developers, including data preprocessing, hyper-parameter selection, and model evaluation, and compares popular deep learning frameworks like TensorFlow, PyTorch, and Keras. Additionally, it discusses cost considerations and recent advances such as attention mechanisms, capsule networks, and model compression. The study concludes with future directions for CNN research and development. Authors in [26] investigate infrastructure-based object detection and tracking systems for cooperative driving automation. It discusses various techniques, including background filtering methods, clustering algorithms like DBSCAN, neural network-based classification, Kalman filter tracking, grid-based detection, attention-based object detection, and multi-sensor fusion techniques. A survey study present in [27] provides a broad overview of the recent development in deep-learning-based object detection in overhead imagery (e.g., drones, airplanes and satellites). It also points out the large-scale real-life uses of such technology, which include crop yield measurements, disaster monitoring, and supply chain forecasting. It also covers the distinct challenges associated with overhead imagery, namely huge image sizes, different image resolutions, small objects and complicated backgrounds. The article examines and contrasts the state-of-the-art detection techniques that are specific to these challenges and presents the newly created overhead image databases.

While some research papers provide general reviews of deep learning-based object detection models like in [28], and discuss their application in autonomous vehicles [21], they pay no special attention to the detection of abandoned objects. Yet, these articles emphasize the significance of such advanced object detection models as R-CNN, Faster R-CNN, YOLO (YOLOv3, YOLOv4, and YOLOR), and 3D convolutional networks that can be used to detect various objects in various areas. In contrast to the methods mentioned above, the proposed framework in this paper has two stages of filtering and detection: an appearance-based CNN pre-filter, which eliminates non-suspicious proposals at the earliest stage, and later YOLOv9 detection with Kalman filter-based time tracking to verify abandonment of objects. The approach ensures minimal computational cost, at high precision and recall, and supports real time inference of 30 FPS in challenging surveillance applications with occlusion, variation in lighting, and background clutters.

III. DATASETS DESCRIPTION

Machine Learning pipelines start with data. In this research work we integrate and utilize two well known datasets in the realm of abandoned object detection namely Performance Evaluation of Tracking and Surveillance (PETS 2006) [29], and Abandoned Object Detection and Analysis (ABODA) [30]. PETS 2006 is a collection of seven video sequences, shot at 25 FPS and 768 × 576 pixels resolution. These videos are recorded in a parking lot in an outdoor location, at various camera angles capturing abandoned luggage. The videos have elaborate bounding box annotations, with abandonment event labels such as left luggage, removed objects and vehicle movement. This makes it a very good dataset for object detection and tracking in a dynamic environment. ABODA on the other hand has 11 video sequences of real-world CCTV surveillance, consisting of extremely challenging environments, such as crowded areas, abrupt illumination changes, night scenarios, and indoors and outdoors conditions. Bounding boxes are carefully placed around abandoned items and ownership noted in each video. Spatial features, and items that are actually abandoned and those that are temporarily discarded, are identified. Combining both of these datasets results in a more diverse and dynamic dataset that covers both controlled and real-world surveillance conditions, making it the ideal dataset to benchmark abandoned object detection algorithms. The multi-camera views together with different lighting scenarios and ownership tags make sure that the data becomes a consistent baseline to test machine learning and deep learning algorithms in the real-life security and surveillance scenarios.

A. DATA PREPROCESSING

Data pre-processing is an essential step in ML pipelines and usually serves two purposes. First is required to transform the input data in such a way as to make it suitable for the input of the ML models. Secondly, it can potentially increase the quality of the input data for efficient training of the ML models and to obtain good performance. In this research work, the following preprocessing steps were utilized.

1) **Frame Extraction:** Frame extraction is performed to retain temporally informative frames while discarding redundant data that does not contribute to abandoned object detection. The process is fully automatic and does not involve any manual frame selection. This process can be mathematically modelled as follows. Let $V = \{F_1, F_2, \ldots, F_T\}$ represent all frames in the original video of duration T seconds with a frame rate of r (e.g., 30 FPS), then:

$$|V| = r \cdot T \tag{1}$$

To reduce temporal redundancy while preserving motion cues, we uniformly sample frames at a fixed sampling rate $r_s = 15$ FPS effectively selecting a frame every 0.25 seconds. This value was empirically selected as a balance between temporal resolution and computational efficiency, as abandoned object events evolve slowly over time and do not require full-frame-rate processing. The sampled frame set $S \subseteq V$ is:

$$S = \{F_i \in V \mid i \equiv 0 \ (\mathrm{mod}\ \lfloor r/r_s \rfloor)\} \tag{2}$$

Hence, the number of sampled frames is approximately:

$$|S| \approx r_s \cdot T \tag{3}$$

Next, an automatic relevance filtering step is applied to eliminate frames that do not contain useful visual information for detection (e.g., empty scenes or frames without foreground activity). A relevance function $\mathcal{R}(F_i) \in \{0, 1\}$ is defined as:

$$\mathcal{R}(F_i) = \begin{cases} 1 & \text{if } F_i \text{ is relevant for the model} \\ 0 & \text{otherwise} \end{cases} \tag{4}$$

The final frame set provided to the CNN–YOLOv9 pipeline is:

$$X = \{F_i \in S \mid \mathcal{R}(F_i) = 1\} \tag{5}$$

This approach reduces computational complexity from:

$$\mathcal{C}_{\text{full}} \propto |V| \tag{6}$$

to

$$\mathcal{C}_{\text{reduced}} \propto |X| \ll |V| \tag{7}$$

Additionally, reducing redundant frames lowers the risk of overfitting, as the model is trained on informative and diverse samples only. The overfitting risk $\mathcal{O}$ is inversely proportional to the number of relevant frames:

$$\mathcal{O} \propto \frac{d}{|X|} \tag{8}$$

where d is the model's capacity (e.g., number of parameters).

2) **Frame Normalization:** For data consistency and comparability, normalization is used to normalize the pixel intensities of the input image. Normalizing frames can potentially enhance the performance and reliability of the model by rescaling abnormally high or low values. The general formula of normalization is represented in the following equation 9.

$$I_{norm}(x, y) = \frac{I(x, y) - I_{min}}{I_{max} - I_{min}} \tag{9}$$

where $I_{norm(x,y)}$ represents the normalized brightness of a pixel. I_{min} is the minimum brightness found in the frame, and I_{max} is the maximum brightness intensity of the frame.

3) **Frame Cropping** Cropping is the process of removing irrelevant areas of an image. We cropped all input images to 512 × 512 pixels. This helps the model to pay attention to more important details. Secondly, it reduces the amount of data in an image and therefore reduces the overall computing cost of the model. Mathematically, this process is expressed as follows.

$$C = I[x_1 : x_2,\ y_1 : y_2] \tag{10}$$

where $I[x_1 : x_2, y_1 : y_2]$ represents a cropped sub-region of the original frame. x_1 and y_1 specify the starting coordinates (top-left corner) while x_2 and y_2 specify the ending coordinates (bottom-right corner) of the cropped region.

4) **Frame Augmentation** There are numerous augmentation techniques, including rotation, scaling, translation, flipping and adjustment of brightness. They render the model robust to changes in real-life conditions, including breaking of illumination, alteration in perspective and object orientations. It is also useful in preventing overfitting by increasing the size of the dataset, thus enhancing the capability of the model to extrapolate on the data. Data Augmentation enhances performance of the models particularly when the datasets are limited in diversity and in frequency. Through augmentation, including Mosaic, Color Jitter, Random Crop, Flip, and MixUp during training with YOLOv9, models can be highly robust to the real-world problems of lighting variations, occlusions, and background clutter. Experimental findings indicate that augmentation is able to improve precision and recall in the two datasets. Specifically, the ABODA dataset is the most benefiting as it consists of more complex and realistic surveillance scenarios. The general formula of augmentation is provided as follows in Equation 11.

$$A(I) = T(I, p_1, p_2, \ldots.., p_n) \tag{11}$$

where $A(I)$ represents the augmented version of the input data I. T denotes a data transformation function. $p_1, p_2, \ldots.., p_n$ are parameters that control the specific augmentation techniques applied, such as rotation angles, scaling factors, translation distances, brightness adjustments, etc.

B. EXPLORATORY DATA ANALYSIS

Results from our exploratory data analysis of the combined dataset are presented in Figure 1 and Figure 2. The bar chart in the top-left of Figure 1 displays the distribution of classes and compares the amount of instances that are labeled as Non-suspicious and Suspicious (abandoned) in the dataset. The distribution seems more or less balanced, as the number of non-suspicious samples are slightly higher. This reduces the problem of imbalance on classes during model training and bias towards any particular class. The bounding box overlay map is depicted in the visualization on the Top-Right of the same figure. It maps all bounding boxes of objects of the dataset onto a normalized image space and visualizes them. Central concentration implies that the majority of the abandoned and non-abandoned items are in the middle part of the field of view, which approximates the systems of a public surveillance (e.g., entrances, hallways). This assists in knowing the location in the scene where objects are supposed to be found in the model. The bottom left plot of this picture is a 2D heatmap of object centres. In this plot, the centroid values of all bounding boxes on the X and Y axis are plotted in 2D. The common spatial locations of objects in the two classes are represented in high density regions. These aid to our methodology (spatio-temporal modeling e.g. heatmaps and positional priors). Finally, the distribution of the bounding box size is presented in a scatter histogram in the Bottom-Right. This histogram represents the co-distribution of width of the bounding box and height. The thick grouping in the bottom section of this picture indicates that the majority of the items are comparatively small (e.g., bags, packages). This can be applied to size optimization of anchor boxes of YOLOv9.

Figure 2 shows a correlogram that provides pairwise comparisons among four key features: x, y, width, and height representing bounding box geometry. Each diagonal panel shows a histogram of the distribution of individual features. Features like width and height are positively skewed, indicating most bounding boxes are small in scale. The x and y distributions show peaks around particular regions, supporting the spatial prior assumption. Each cell in the lower triangle displays a 2D histogram of two features (e.g., x vs. y, width vs. height). The x-y plot confirms spatial clustering, while the width-height plot again shows small object size dominance. The plots reveal how certain features co-vary, essential for understanding object shape consistency.

As shown in Figure 2 and numerically summarized in Table 2, the width and height of bounding boxes show a strong positive correlation (≈ 0.72). This indicates that most

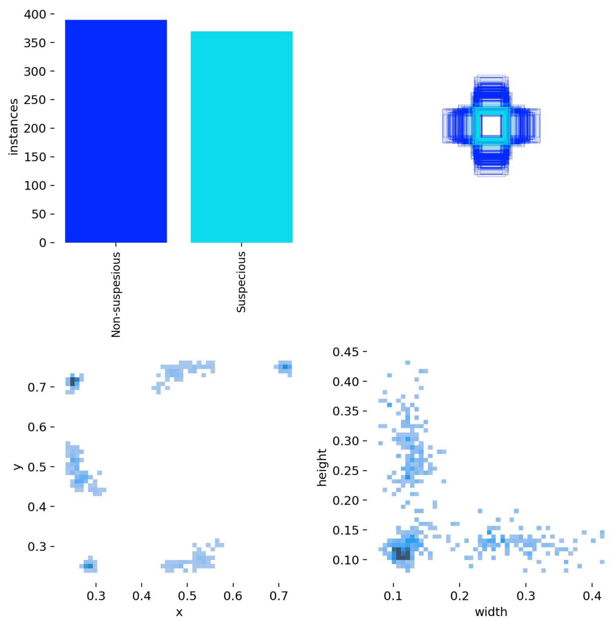


FIGURE 1. Exploratory Data Analysis of the combined PETS2006 and ABODA dataset showing (Top-left) Class distribution,(Top-right) Spatial overlay of bounding boxes,(Bottom-left) Centroid heatmaps.(Bottom-right) Size distribution plots.

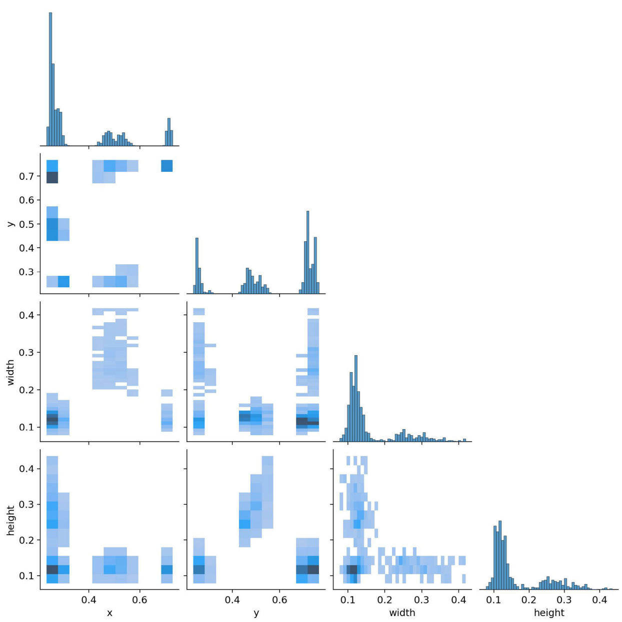


FIGURE 2. Pairwise correlation heatmap and feature distribution matrix across spatial (x, y) and shape (width, height) features.

objects in the dataset have consistent aspect ratios, which are characteristic of common abandoned items such as bags or containers. Conversely, the spatial coordinates (x, y) show minimum correlation with object dimensions, suggesting that objects appear across a wide area without being constrained to specific sizes based on location. These insights support the

TABLE 2. Approximate pearson correlation matrix among bounding box features, estimated from visual analysis of the dataset.

	x	y	width	height
x	1.00	0.02	0.15	0.12
y	0.02	1.00	0.05	0.09
width	0.15	0.05	1.00	0.72
height	0.12	0.09	0.72	1.00

use of location-independent anchor design and validate the inclusion of both spatial and shape features in our detection pipeline.

IV. METHODOLOGY

Abandoned object detection in the proposed system begins with data acquisition, wherein surveillance footage or curated video datasets containing unattended object scenarios are collected. The overall detection process, illustrated in Figure 3 and Algorithm 1, comprises three main components: (1) a robust preprocessing stage that enhances data quality through frame extraction, resizing, noise reduction, and normalization, as described in the previous section; (2) the CNN stage, which operates solely on candidate object regions extracted from individual frames, performing appearance-based binary classification into *suspicious* and *non-suspicious* categories; and (3) the YOLOv9 architecture, augmented with Kalman filtering, which receives only those regions flagged as

Algorithm 1 Two-Stage Abandoned Object Detection Framework

Require: Video frames $f_1, \ldots, f_n$
Ensure: List of detected abandoned objects with bounding boxes, labels, and timestamps

```
for each frame f_t do
    Preprocess f_t (frame extraction, resizing, noise reduction, normalization)
    Extract object proposals {o_i}
    for each object o_i in {o_i} do
        C_CNN(o_i) ← CNN_classify(o_i)
    end for
    S ← {o_i | C_CNN(o_i) ≥ τ_app}        ▷ forward suspicious only
    for each object o_i in S do
        C_YOLO_KF(o_i) ← YOLOv9_detect_and_track_CNN_flagged(o_i)    ▷ Run YOLOv9 with Kalman tracking only on objects flagged by CNN
        C_final(o_i) ← α·C_YOLO_KF(o_i) + (1−α)·C_CNN(o_i)
        if C_final(o_i) ≥ θ then
            Trigger alarm with bounding box, classification label, timestamp
        end if
    end for
end for
```

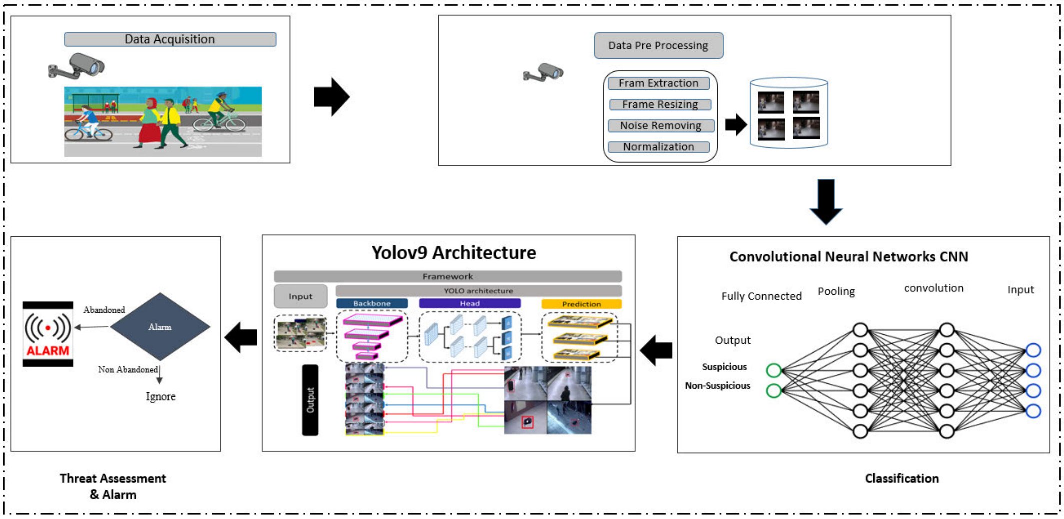


FIGURE 3. **Proposed two-stage framework for abandoned object detection. In Stage 1, the CNN performs appearance-based classification on object proposals, forwarding only those flagged as suspicious. In Stage 2, YOLOv9 detects and tracks these CNN-flagged objects over time using Kalman filtering to assess stationarity. A fusion of CNN appearance confidence and YOLOv9+Kalman spatiotemporal confidence determines abandonment, triggering an alert if the threshold is exceeded.**

suspicious by the CNN and performs precise spatial localization and temporal tracking of these objects. YOLOv9 is selected as the localization backbone due to its single-stage architecture, which enables real-time inference with stable bounding box predictions required for continuous video surveillance. Its efficiency and temporal consistency make it well-suited for Kalman filter–based tracking. Moreover, YOLOv9 demonstrates strong performance in detecting small and medium-sized objects, which is critical for abandoned object detection. In the proposed framework, YOLOv9 is used exclusively for spatial localization, while appearance-based discrimination is handled separately by the CNN module. The decision on whether an object is abandoned is made in the final fusion stage, which combines the CNN's appearance score with YOLOv9 + Kalman's spatiotemporal confidence. An object is classified as abandoned if it remains stationary beyond a predefined temporal threshold and lacks an associated person. Upon detection, an alarm is triggered to alert security personnel in real time. This integrated approach balances real-time detection accuracy with contextual awareness by leveraging appearance-based classification and temporal persistence.

A. SUSPICIOUS OBJECT CLASSIFICATION USING CNN

Following preprocessing, object regions obtained from frame-level segmentation are passed through a CNN to determine whether they exhibit visual characteristics commonly associated with suspicious items. The CNN performs appearance-based classification using spatial features such as shape, texture, size, and contour complexity, producing an appearance confidence score $C_{\mathrm{CNN}}(o_i)$ for each candidate region o_i. Only objects classified as *suspicious* (i.e., $C_{\mathrm{CNN}}(o_i) \geq \tau_{\mathrm{app}}$) are forwarded to the YOLOv9+Kalman tracking module for subsequent spatiotemporal analysis. This early filtering step reduces computational load by discarding visually irrelevant or non-threatening objects, allowing the system to focus its resources on high-risk candidates. The CNN is trained on a labeled dataset containing examples of typical suspicious items in public spaces (e.g., unattended bags, boxes, or packages). Feature extraction may also incorporate contextual cues (e.g., proximity to entry points or restricted zones), although the final determination of whether an object is abandoned is made at the alarm generation stage.

Only objects labeled as *suspicious* are passed to the YOLOv9-based tracking module for further spatial and temporal analysis. The cost reduction from this filtering can be estimated as follows:

Let o_i represent an object extracted from frame f_t. Each object is transformed into a fixed-size feature map $F(o_i)$, which is input to the CNN classifier $\mathcal{C}$. The output of the classifier is a label:

$$\hat{y}_i = \mathcal{C}(F(o_i)) \in \{\text{suspicious, non-suspicious}\} \tag{12}$$

The computational efficiency improvement from this early filtering can be estimated as follows. Let N be the total number of detected objects, and let $\alpha \in [0, 1]$ denote the proportion classified as *suspicious* by the CNN. Let C_{CNN} and C_{YOLO} represent the average computational cost per object for CNN classification and YOLOv9 processing, respectively.

Without filtering:

$$\mathcal{C}_{\mathrm{full}} = N \cdot (C_{\mathrm{CNN}} + C_{\mathrm{YOLO}}) \tag{13}$$

With filtering:

$$\mathcal{C}_{\mathrm{filtered}} = N \cdot C_{\mathrm{CNN}} + \alpha N \cdot C_{\mathrm{YOLO}} \tag{14}$$

Relative cost reduction ratio (RR):

$$\mathrm{RR} = \frac{\mathcal{C}_{\mathrm{full}} - \mathcal{C}_{\mathrm{filtered}}}{\mathcal{C}_{\mathrm{full}}} = \frac{(1-\alpha) \cdot C_{\mathrm{YOLO}}}{C_{\mathrm{CNN}} + C_{\mathrm{YOLO}}} \tag{15}$$

B. SPATIOTEMPORAL TRACKING WITH YOLOV9 AND KALMAN FILTERING

Entities that CNN labels as suspicious are sent to a detection and tracking framework based on the CNN model (YOLOv9) to analyze the objects. The functions of this stage are twofold (1) accurate spatial localization of the suspicious object on a frame-by-frame basis, and (2) time tracking to establish whether the object is at rest or in motion.

YOLOv9 has two main architectural improvements, namely Programmable Gradient Information (PGI) which preserves fine object details in training and Generalized Efficient Layer Aggregation Network (GELAN) which improves the efficiency of the backbone. PGI is more useful in identifying small or partially covered objects whereas GELAN is accurate with limited calculation cost. The pre-trained YOLOv9 model was rendered in fine-tuning on benchmark PETS2006 and ABODA by the transfer learning approach and improved the performance in various conditions, such as low light and occlusion. This adaptation enhanced the small object and partially occluded mean Average Precision (mAP). YOLO was selected as the core detection backbone due to its favorable balance between detection accuracy and real-time performance, which is essential for abandoned object surveillance. As a single-stage detector, YOLO avoids region proposal overhead and enables fast inference on continuous video streams. Furthermore, recent YOLO versions demonstrate strong performance in small-object detection through multi-scale feature aggregation, making them well suited for identifying abandoned objects such as bags or parcels. The stable and consistent bounding box predictions produced by YOLO also facilitate reliable Kalman filter–based tracking and temporal stationarity analysis. These characteristics make YOLO an effective and practical choice for the proposed spatiotemporal framework.

Following detection, Kalman filtering is applied to maintain object identity across frames and estimate motion. Let $B_t(o_i)$ denote the bounding box of object o_i at frame t, and $d(B_t, B_{t-1})$ be the Euclidean distance between bounding box centroids in consecutive frames. An object is considered stationary in frame t if:

$$\Delta_t(o_i) = \begin{cases} 1 & \text{if } d(B_t(o_i), B_{t-1}(o_i)) \leq \epsilon \\ 0 & \text{otherwise} \end{cases} \tag{16}$$

The stationary duration is:

$$T_{\text{stationary}}(o_i) = \sum_{t=t_0}^{t_n} \Delta_t(o_i) \tag{17}$$

If the stationary duration exceeds a predefined temporal threshold τ, the object is considered abandoned. With a sampling rate of 15 FPS, τ is set to 75 consecutive stationary frames (approximately 5 seconds), although in real-world deployments it might be a configurable parameter. Mathematically

$$\delta(o_i) = \begin{cases} 1 & \text{if } T_{\text{stationary}}(o_i) \geq \tau \\ 0 & \text{otherwise} \end{cases} \tag{18}$$

C. ABANDONMENT DECISION AND ALARM GENERATION

To reduce false positives and incorporate both appearance and temporal context, the final decision on whether an object is abandoned uses a weighted fusion of CNN and YOLOv9 + Kalman outputs, given as follows.

- $C_{\text{CNN}}(o_i)$: appearance confidence score from CNN classification.
- $C_{\text{YOLO+KF}}(o_i)$: spatiotemporal confidence score from YOLOv9 detection with Kalman tracking.

The final abandonment confidence is:

$$C_{\text{final}}(o_i) = \alpha \cdot C_{\text{YOLO+KF}}(o_i) + (1 - \alpha) \cdot C_{\text{CNN}}(o_i) \tag{19}$$

where $\alpha \in [0, 1]$ balances spatial-temporal evidence and appearance classification.

An object is declared abandoned if:

$$C_{\text{final}}(o_i) \geq \theta \tag{20}$$

where $\theta \in [0, 1]$ is a decision threshold tuned for optimal precision–recall balance.

1) ALARM TRIGGERING

If the final abandonment confidence exceeds θ i.e. $C_{\text{final}}(o_i) \geq \theta$, the system triggers an alert containing:

- Bounding box coordinates from YOLOv9.
- Suspicious classification result from CNN.
- Timestamp and location within the video frame.

This structured decision process by fusing appearance and spatiotemporal evidence ensures that only high-confidence abandoned objects trigger alarms, minimizing unnecessary alerts (false positives) and high detection sensitivity, enabling reliable operation in real-world public surveillance deployments.

D. EVALUATION METRICS

The performance of the proposed methodology is evaluated using specific performance measures, including accuracy, precision, and recall.

1) **Accuracy:** Accuracy is a metric commonly employed to provide an overall assessment of a model's performance across all classes. This metric is particularly valuable when all classes hold equal significance, quantifying the correctness of predictions by dividing the number of accurate predictions by the total number of predictions; the following equation demonstrates accuracy.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{21}$$

2) **Precision:** Precision is determined by dividing the number of Positive samples correctly classified by the

total count of samples classified as Positive, whether they were classified correctly or not. It serves as an indicator of the model's accuracy in classifying samples as positive, specifically measuring its ability to correctly identify positive instances. Precision is presented by the following equation.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{22}$$

3) **Recall:** Recall is computed by dividing the number of correctly classified Positive samples by the total count of Positive samples. It quantifies the model's capability to identify Positive samples accurately, essentially gauging its sensitivity in detecting such instances. A higher recall value signifies a greater ability to detect Positive samples within the dataset. Recall is illustrated in the following equation.

$$\text{Recall} = \frac{TP}{TP + FN} \tag{23}$$

where True Positive (TP): The model correctly identifies something as positive, and it is indeed positive. True Negative (TN): The model correctly recognizes something negative, and it is indeed negative. False Positive (FP): The model mistakenly identifies something as positive when it's negative. False Negative (FN): The model erroneously identifies something as negative when it's positive.

V. EXPERIMENTAL SETUP, RESULTS, AND DISCUSSION

In this section, first, we present the implementation details and then an empirical evaluation of our proposed model. In empirical analysis, first, we look into the performance of CNN for the classification of objects into suspicious and non-suspicious classes. Then, we evaluate the performance of the object detection models Yolo-9 and its variants. Finally, we compare the effectiveness of our proposed model with other state-of-the-art models in the field. This comparative analysis offers valuable insights into the model's performance and its standing in the broader research landscape.

A. EXPERIMENTAL SETUP

The datasets were analyzed and evaluated by the experimental approach involving the exploratory analysis presented in Section III. In order to maximize computational efficiency, but still provide temporal coherence, the raw video streams were downsampled to 15 FPS before processing. The resultant reduction of the per-frame computational demand does not corrupt the adequate temporal granularity to detect objects accurately and discard them appropriately.

It was implemented in Python version 3.10.4, PyTorch version 1.12.1, and CUDA version 11.7 and trained and inferred with the help of the powerful 24 GB VRAM NVIDIA GeForce RTX 3090 GPU as provided by the Google Colab. This execution environment was a cloud environment with a smooth integration to the use of GPU acceleration, so training and evaluation could occur effectively without having to use a dedicated local machine.

Training was done on models on a total of 100 epochs. A batch size of 64 was used to strike a balance between training performance and the use of GPU memory, and this was the largest batch size that did not overflow the memory of the RTX 3090 (using 24 GB of VRAM). The binary classification was optimized byBinary Cross-Entropy loss to train the binary classification. During preprocessing, data was sorted and labeled to enable the model to detect static objects.

Datasets were well structured and labeled with bounding boxes around objects of interest to help in the learning process, which constitutes the most important aspect of the CNN classification and YOLOv9 detection. The dataset included the changes in lighting, background and object positions which mirror the real-world issues of abandoned object recognition. In order to validate the model, the datasets were divided into 70 percent training, 15 percent validation, and 15 percent testing sets of both PETS2006 and ABODA datasets. Among the validation sets, 1000 labeled images were to be closely evaluated in terms of performance.

The metrics that were used to monitor the training process included accuracy, F1-score, precision, recall, and mean Average Precision (mAP). Elements of losses, such as box loss, classification loss, and distribution focal loss, were monitored in training and validation so as to have a stable convergence and to reduce overfitting.

B. PERFORMANCE EVALUATION OF CNN

The CNN classifier achieved an accuracy of 99.35% and an F1-score of 99.05% on the held-out 15% test split, demonstrating its strong capability in capturing spatial features from candidate object regions extracted during preprocessing. The confusion matrix in Figure 4 confirms that the model effectively differentiates between suspicious and non-suspicious objects with minimal misclassification. These results underscore the suitability of CNNs for appearance-based filtering in the proposed two-stage framework.

The proposed classifier model's performance can be effectively illustrated through a set of learning graphs, as shown in Figure 5, that highlight its behavior across different metrics and confidence thresholds for the two classes: suspicious and non-suspicious. Firstly, the precision-recall curve Figure 5(a) provides a visual representation of the trade-off between precision and recall, indicating how well the model maintains accuracy in identifying true positives while minimizing false positives for both classes. This is particularly important in scenarios with class imbalance. Secondly, the precision-confidence curve Figure 5(b) shows how precision varies with respect to the prediction confidence score. This curve helps in understanding how trustworthy the model's predictions are at various confidence levels and assists in selecting an optimal decision threshold. Thirdly, the recall-confidence curve in Figure 5(c) illustrates the change in recall as the confidence threshold shifts, highlighting the model's sensitivity and its ability to capture all relevant instances.

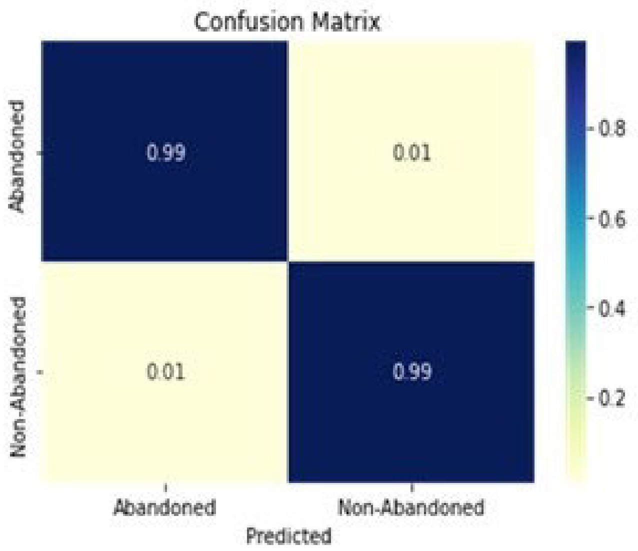
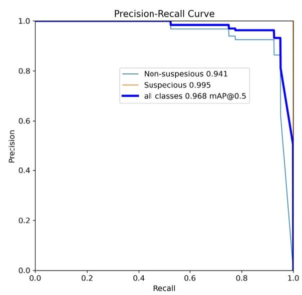
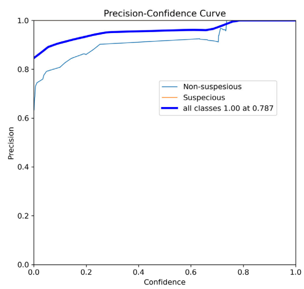


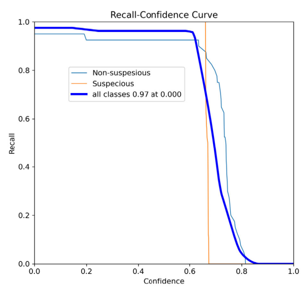
FIGURE 4. **Convolution Neural Network Confusion Matrix.**



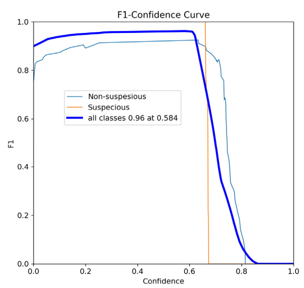
(a) PR curve



(b) P curve



(c) Recall



(d) F1 curve

FIGURE 5. **Proposed Classifier model learning graphs:(a) precision-recall curve,(b) precision-confidence curve,(c) recall-confidence curve,(d) F1-confidence curve for suspicious and non-suspicious classes.**

Lastly, the F1-confidence curve Figure 5(d) combines both precision and recall into a single performance metric and plots it against confidence scores, offering insights into the balance between the two and aiding in determining the most effective confidence threshold for classification. Together, these graphs provide a comprehensive view of the classifier's performance for both suspicious and non-suspicious classes.

C. OBJECT DETECTION PERFORMANCE

In this section, we report the object detection performance of YOLOv9. The first step is to detect the object of interest, which in our case is a potentially abandoned object in a scene. Figure 6 illustrates precise bounding box predictions of YOLOv9 around suspicious objects in a variety of environments. These results demonstrate the YOLOv9 localizer's robustness and accuracy, benefiting from the balanced nature of the dataset, which ensures fair representation across different object types and scene complexities.

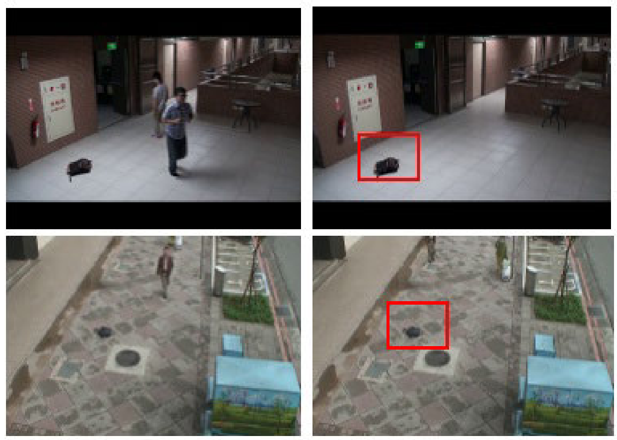


FIGURE 6. **Example results of suspicious object localization on the ABODA dataset using the proposed YOLOv9 localizer, showing high accuracy across varied backgrounds.**

1) LOCALIZATION LOSS AND TRAINING PROCEDURE

The proposed localizer is implemented using YOLOv9 and follows its standard multi-component loss formulation. Specifically, the localization loss is composed of three terms: (i) a bounding box regression loss based on Intersection-over-Union (IoU) optimization to ensure accurate spatial localization, (ii) an objectness confidence loss that estimates the likelihood of an object being present in a predicted region, and (iii) a classification loss that penalizes incorrect class predictions. These components are jointly optimized during training to achieve robust object localization performance.

The YOLOv9 model is trained end-to-end using annotated bounding box data, while the CNN-based appearance classifier is trained separately for suspicious object recognition. No modifications are made to the original YOLOv9 loss formulation. During inference, the CNN acts as an appearance-based filter, and YOLOv9 performs spatial detection and Kalman filter–based temporal tracking. The abandoned object decision is determined using a temporal persistence threshold τ, which operates independently of the YOLOv9 training loss.

a: TRAINING AND VALIDATION BEHAVIOR

The proposed localizer model was trained and validated using a comprehensive loss monitoring strategy. During training, the bounding box loss, classification loss, and distribution focal loss were jointly minimized, exhibiting stable convergence over 100 epochs, as shown in Figure 7(a). Model performance was evaluated using bounding-box-based metrics (Figure 7(b)), where both precision and recall consistently improved throughout training.

Figure 7(c) presents the validation box loss, classification loss, and distribution focal loss, indicating effective generalization without noticeable overfitting. Furthermore, Figure 7(d) shows a steady increase in mAP_{50} and $mAP_{50\text{-}95}$ on the validation set, confirming improved object localization accuracy. These results demonstrate the robustness and efficiency of the proposed localizer for real-time object detection tasks.

b: COMPARISON WITH YOLOv9 VARIANTS

Table 3 compares the performance of five YOLOv9 variants — YOLOv9t, YOLOv9s, YOLOv9m, YOLOv9c, and the proposed YOLOv9e — across three evaluation metrics: precision, recall, and F1-score. The proposed YOLOv9e achieves the highest overall performance, with precision and recall both exceeding 0.99, leading to an F1-score near 0.99. In contrast, lighter variants such as YOLOv9t and YOLOv9s demonstrate reduced accuracy, particularly in terms of F1-score, highlighting the trade-off between model complexity and detection accuracy.

TABLE 3. Object detection performance of YOLOv9 variants.

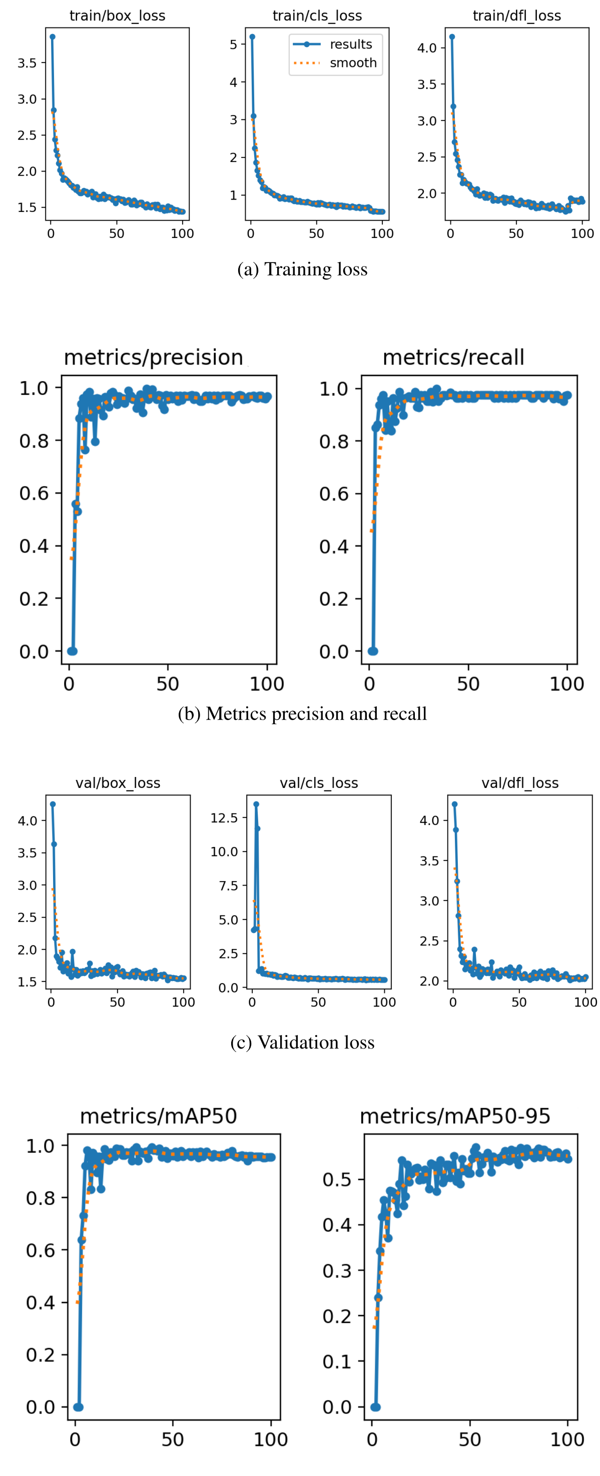
Model	Precision	Recall	F1-Score
YOLOv9t	0.9450	0.9326	0.9388
YOLOv9s	0.9736	0.9880	0.9807
YOLOv9m	0.9860	0.9560	0.9708
YOLOv9c	0.9736	0.9765	0.9751
YOLOv9e (Proposed)	0.9981	0.9976	0.9901

2) COMPUTATIONAL COST

Table 4 and Figure 8 present a comparative analysis of computational characteristics for the five YOLOv9 variants. As expected, the resource footprint increases with model complexity. YOLOv9t and YOLOv9s are lightweight models with minimal computational demand, suitable for real-time applications on edge devices. In contrast, YOLOv9e consumes more resources but provides unmatched detection accuracy, making it well-suited for high-stakes surveillance scenarios.

a: INTERPRETATION AND INSIGHTS

Better architecture, better feature aggregation with GELAN, and better gradient flow with PGI can be associated with the superior performance of YOLOv9e. Furthermore, the model can process CNN-flagged suspicious regions only which reduces unnecessary detections and increases speed and accuracy. Kalman filtering further makes object tracking smoother to prevent false alarms produced by the momentary occlusions or the background motion. Although YOLOv9e is computationally more expensive than lightweight versions, in the context of the critical security, the trade-off is warranted by the near-flawless detection metrics and strong real-time performance.



(a) Training loss

(b) Metrics precision and recall

(c) Validation loss

(d) Metrics mAP

FIGURE 7. Proposed Localizer model training and validation loss graphs:(a) training box/classification/distribution focal loss,(b) metrics precision and recall,(c) validation box/classification/distribution focal loss,(d) metrics mAP50 and mAP50-95.

TABLE 4. Comparison of YOLOv9 variants in terms of model size, memory, speed, and computational cost.

Model	Parameters (M)	Memory Usage (MB)	Inference Speed (ms)	GFLOPs
YOLOv9t	2.0	4.6	5.0	7.6
YOLOv9s	7.1	15.2	2.8	26.7
YOLOv9m	20.0	40.8	9.9	76.5
YOLOv9c	25.3	51.6	16.9	103.7
YOLOv9e (Proposed)	57.3	117.2	10.0	189.1

TABLE 5. Comparison of the proposed localizer with State-of-the-Art (SOTA).

Author	Accuracy (%)	Precision (%)	Recall (%)
Mahalingam [31]	65.00	71.00	65.00
Lwin and Tun [32]	84.55	98.80	97.70
Ramachandran [33]	95.15	98.80	98.40
Arbab M Qasim [34]	99.00	99.00	99.00
Teja, YD [11]	99.00	91.00	88.00
Proposed	**99.81**	**99.67**	**99.01**

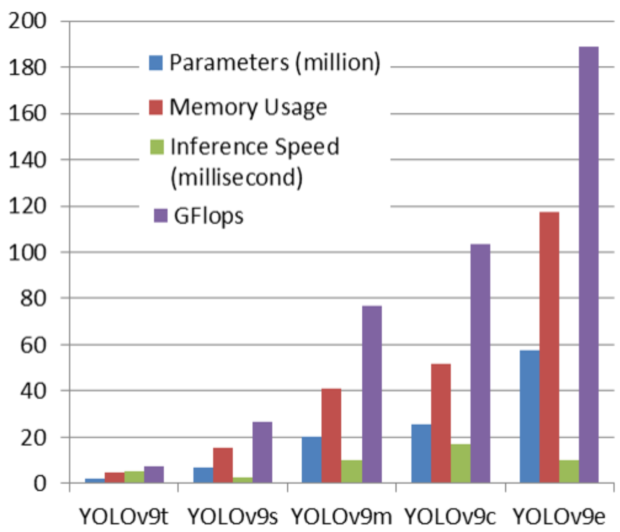


FIGURE 8. Comparison of YOLOv9 Variants in Terms of Model Size, Memory, Speed, and Computational Cost.

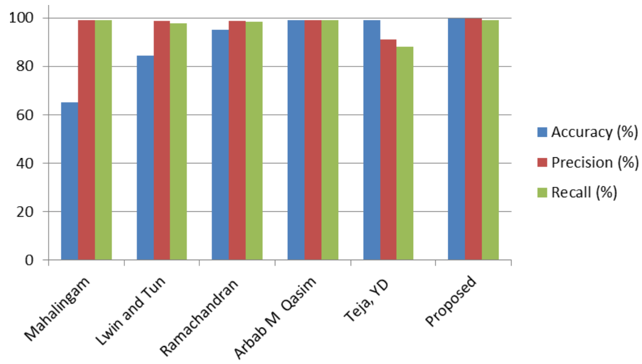


FIGURE 9. Comparison of the Proposed Localizer with State-of-the-Art (SOTA).

D. COMPARISON WITH STATE OF THE ART (SOTA)

Table 5 and Figure 9 provide a comparison between the proposed abandoned object detection method and some of the existing methods in recent literature, such as Mahalingam, Lwin and Tun, Ramachandran, Arbab M. Qasim and Teja Y.D. It has been evaluated using three metrics, which are accuracy, precision and recall. All three metrics provide a score of greater than 99 percent in the proposed model which means that the model is better in detecting and classifying abandoned objects correctly. Although other methods that have been developed by Arbab M. Qasim and Ramachandran also achieve a high level of recall and precision, their accuracy is slightly lower. Previous literature, including that provided by Mahalingam, exhibits much worse accuracy and poor metric performance. This comparison justifies efficiency and strength of the proposed approach to real-life surveillance.

Interpretation and Insights: Our design has made several decisions that could be attributed to notably better performance compared to existing works:

Two-stage pipeline — The system can ensure that only the relevant objects are processed by YOLov9 by first-filtering the candidates with the understanding of a CNN classifier to minimize false positives and focus more on detection.

Kalman filtering using YOLOv9e — This is the best option as it provides better localization and temporal stability, which is important in detecting objects that have not been attended to in some time.

Well-balanced datasets and powerful preprocessing - The balanced presence of object categories in the dataset avoids discrimination and allows the detector to extrapolate well to various real-life scenes.

In comparison with the single-stage detectors in earlier research, our approach is able to deal with the complexity of the scene and dynamic scenes, as can be seen by the high accuracy, precision, and recall. The narrow gap between the performance of precision and recall also means that the model is successful in striking a balance between false alarms and minimizing false detections, which is an important aspect of high-stakes surveillance applications.

VI. CONCLUSION

This paper presents a two-stage spatiotemporal CNN–YOLOv9 framework for high-precision, real-time abandoned object detection in public surveillance videos. The framework integrates CNN-based classification to identify suspicious objects with YOLOv9 spatial localization, enhanced by a Kalman filter for robust tracking, and uses a temporal threshold (τ) to detect abandoned objects accurately. This combination significantly reduces false positives while maintaining high detection sensitivity. Extensive experiments on the PETS2006 and ABODA datasets demonstrate that the proposed framework achieves 99.81% accuracy, 99.67% precision, and 99.01% recall, outperforming previous state-of-the-art methods. These results confirm that the system provides reliable object detection and localization, with high robustness and real-time performance suitable for deployment in high-stakes public environments such as airports, train stations, and shopping centers. The framework's modular design allows integration with future YOLO variants, additional sensor data, or edge devices, enhancing its adaptability to evolving security requirements. Despite its high performance, the system remains sensitive to extreme occlusions, challenging lighting and weather conditions, and high computational demands, which may limit deployment on some edge devices. The detection relies on a predefined temporal threshold (τ) for alarm triggering, which may need adjustment for different surveillance settings or scenarios Future work may focus on lightweight YOLO variants, transformer-based feature extraction, low-light and occlusion handling, and efficient edge deployment, improving real-time processing, operational reliability, and applicability in diverse scenarios. Overall, the proposed approach provides a robust, scalable, and high-performing solution for automated surveillance, bridging reliability gaps in previous systems and advancing intelligent security monitoring in public spaces.

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